

Type 2 diabetes mellitus after gestational diabetes: a systematic review and meta-analysis.



BACKGROUND: Women with gestational diabetes are at increased risk of developing type 2 diabetes, but the risk and time of onset have not been fully quantified. We therefore did a comprehensive systematic review and meta-analysis to assess the strength of association between these conditions and the effect of factors that might modify the risk. **METHODS:** We identified cohort studies in which women who had developed type 2 diabetes after gestational diabetes were followed up between Jan 1, 1960, and Jan 31, 2009, from Embase and Medline. 205 relevant reports were hand searched. We selected 20 studies that included 675 455 women and 10 859 type 2 diabetic events. We calculated and pooled unadjusted relative risks (RRs) with 95% CIs for each study using a random-effects model. Subgroups analysed were the number of cases of type 2 diabetes, ethnic origin, duration of follow-up, maternal age, body-mass index, and diagnostic criteria. **FINDINGS:** Women with gestational diabetes had an increased risk of developing type 2 diabetes compared with those who had a normoglycaemic pregnancy (RR 7.43, 95% CI 4.79-11.51). Although the largest study (659 164 women; 9502 cases of type 2 diabetes) had the largest RR (12.6, 95% CI 12.15-13.19), RRs were generally consistent among the subgroups assessed. **INTERPRETATION:** Increased awareness of the magnitude and timing of the risk of type 2 diabetes after gestational diabetes among patients and clinicians could provide an opportunity to test and use dietary, lifestyle, and pharmacological interventions that might prevent or delay the onset of type 2 diabetes in affected women. **FUNDING:** None.